

Problem 1

Let

$$J_m(\lambda) = \begin{pmatrix} \lambda & 1 & 0 & \cdots & 0 \\ 0 & \lambda & 1 & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & \lambda & 1 \\ 0 & \cdots & \cdots & 0 & \lambda \end{pmatrix}.$$

Write $J_m(\lambda)^n = (c_{ij}^{(n)})$. Since the only nonzero entries in column j of $J_m(\lambda)$ are λ in row j and 1 in row $j-1$ (when $j > 1$), matrix multiplication gives

$$c_{ij}^{(n+1)} = \lambda c_{ij}^{(n)} + c_{i,j-1}^{(n)},$$

where an index outside $1, \dots, m$ contributes 0.

We claim that, for every $n \geq 0$,

$$c_{ij}^{(n)} = \begin{cases} \binom{n}{j-i} \lambda^{n-(j-i)}, & 0 \leq j-i \leq \min\{n, m-1\}, \\ 0, & \text{otherwise.} \end{cases}$$

For $n = 0$, this is the identity matrix. Assume it holds for n . If $k = j - i$ satisfies $1 \leq k \leq n$, then the recurrence and Pascal's identity give

$$c_{i,i+k}^{(n+1)} = \lambda \binom{n}{k} \lambda^{n-k} + \binom{n}{k-1} \lambda^{n-k+1} = \binom{n+1}{k} \lambda^{n+1-k}.$$

For $k = 0$ the recurrence gives λ^{n+1} ; for $k = n+1$ the second term gives 1; and for $k > n+1$ both terms are zero. Thus the claim holds by induction.

Therefore

$$(J_m(\lambda)^n)_{ij} = \begin{cases} \binom{n}{j-i} \lambda^{n-(j-i)}, & 0 \leq j-i \leq \min\{n, m-1\}, \\ 0, & \text{otherwise.} \end{cases}$$

If $b_k = \binom{n}{k} \lambda^{n-k}$ for $0 \leq k \leq n$ and $b_k = 0$ for $k > n$, this is

$$J_m(\lambda)^n = \begin{pmatrix} b_0 & b_1 & b_2 & \cdots & b_{m-1} \\ 0 & b_0 & b_1 & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & b_2 \\ 0 & \cdots & 0 & b_0 & b_1 \\ 0 & \cdots & \cdots & 0 & b_0 \end{pmatrix}.$$

Problem 2

Let $A = (a_{ij})_{i,j=1}^n$, where $a_{ii} = 2$ and $a_{ij} = (-1)^{|i-j|}$ for $i \neq j$. Multiply row i and column i by $(-1)^{i-1}$ for every i . Equivalently, first multiply each row by these signs and then multiply each

column by the same signs. The determinant is unchanged because the two products of signs square to 1.

Call the resulting matrix B . Its diagonal entries are 2. For $i \neq j$,

$$B_{ij} = (-1)^{i-1}(-1)^{|i-j|}(-1)^{j-1} = 1,$$

because $|i-j|$, $i-j$, and $i+j$ have the same parity. Thus B has 2 on the diagonal and 1 everywhere else.

For $i = 2, \dots, n$, replace row i by row i minus row 1. The first row remains $(2, 1, \dots, 1)$, and each later row becomes

$$(-1, 0, \dots, 0, 1, 0, \dots, 0),$$

with the 1 in column i . Now replace column 1 by the sum of all columns. The resulting matrix is upper triangular: its first diagonal entry is

$$2 - (n-1)(-1) = n+1,$$

and every other diagonal entry is 1. These row and column operations preserve the determinant, so

$\det A = \det B = (n+1) \cdot 1^{n-1} = n+1.$
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